

Diffusion-Guided Weakly Supervised Semantic Segmentation

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Abstract. Weakly Supervised Semantic Segmentation (WSSS) with classification labels typically uses Class Activation Maps to localize the object based on Convolutional Neural Networks (CNN). With limited receptive fields, CNN-based CAMs often fail to localize the whole object. The emergence of a Vision Transformer (ViT) alleviates the problem with superior performance, but the lack of locality in ViT introduces a new challenge. Inspired by the ability of Denoising Diffusion Probabilistic Models (DDPM) to capture high-level semantic information, we bring diffusion models to WSSS to resolve the problem. Firstly, to fuse and semantically align the information between DDPM and ViT, we design the Locality Fusion Cross Attention (LFCA) module. Using the aggregated features from the denoising process of the pretrained DDPM, LFCA generates CAMs (*i.e.* Diffusion-CAMs) that provide locality information to CAMs from ViT (*i.e.* ViT-CAMs). Secondly, by adding noise to the original image and denoising it with DDPM, we obtain a denoised image that can be leveraged as an augmented sample. To effectively guide ViT in excavating the relation between the patches, we devise the Patch Affinity Consistency (PAC) between the outputs of the original image and the denoised image. Extensive ablation studies support the superiority of the proposed method. Our method achieves new state-of-the-art performance on two widely used datasets in WSSS; PASCAL VOC 2012 and MS-COCO 2014. The code is available at <https://github.com/yoon307/DiG>.

Keywords: Denoising Diffusion Probabilistic Models · Weakly Supervised Semantic Segmentation

1 Introduction

To alleviate the labor-intensive and costly annotation procedure in fully-supervised semantic segmentation models, training semantic segmentation with weak labels is proposed. With the inexpensive and readily available labels such as image-level labels [25, 32, 42, 59, 63–65], scribbles [33, 50], and bounding boxes [24, 28], Weakly Supervised Semantic Segmentation (WSSS) methods are actively explored aiming to reach up to full-supervision performance.

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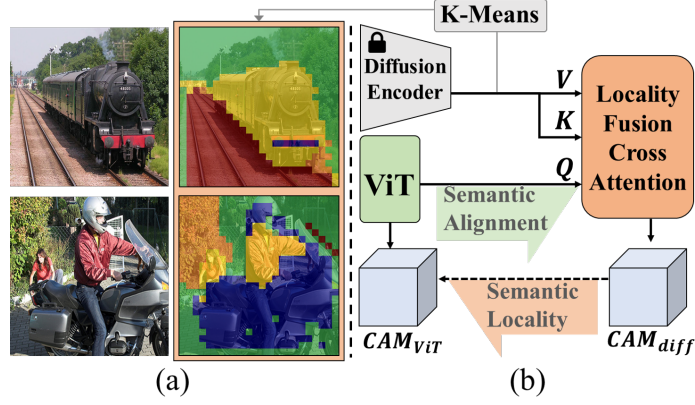


Fig. 1: (a) K-Means clustering result of features from the pre-trained Diffusion model. The same color in the right image indicates that features located in corresponding pixels are in the same cluster. (b) Simplified pipeline of the proposed framework. To solve the locality problem in ViT, the Locality Fusion Cross Attention (LFCA) module provides semantically well-localized Diffusion-CAMs aimed at improving the locality of ViT-CAMs.

Though training a semantic segmentation model with image-level labels is the most practical setting, extracting localization information is challenging since labels only inform the existence of certain objects. To capture spatial information from the classifier and generate pseudo pixel-level labels, most WSSS works utilize Class Activation Maps (CAMs) [67] based on Convolutional Neural Networks (CNNs). However, because of the inherent characteristics of CNNs-based CAMs, they not only put focus on the most discriminative regions (*i.e.*, under-activation) but also have imprecise object boundaries. To facilitate the utilization of acquired CAMs as pseudo-labels for semantic segmentation, various studies have endeavored to improve the quality of CAMs.

Transitioning to CAMs extraction using Vision Transformers (ViT), which harness global features via Multi-Head Self-Attention (MHSA) mechanisms, effectively mitigates the under-activation issue of CNN-based CAMs and shows promising results. However, employing ViTs for CAMs introduces another challenge: ViTs may not exhibit the same level of proficiency in capturing fine-grained local patterns when compared to CNNs. Some approaches alleviate this lack of locality in ViT [19, 31, 48, 54] by improving positional encodings [54] or using hybrid models that combine ViTs with CNNs [19]. However, in WSSS, efforts to address the performance degradation of CAMs due to the lack of locality in ViTs have received relatively less attention.

Meanwhile, Denoising Diffusion Probabilistic Models (DDPM) [21] have shown promising results in various generative modeling tasks such as image generation, inpainting, and super-resolution. It has also been shown that the feature representation from DDPM has the capability to capture high-level semantic informa-

tion [3], valuable for semantic segmentation. Considering that diffusion models can also be trained in a *label-free* manner, it is apt for use in WSSS. From this motivation, we propose a novel WSSS framework that incorporates semantically well-clustered DDPM features with ViT to improve the quality of CAMs. We also propose a self-supervised learning technique that leverages images restored from the DDPM noising and denoising process to refine the quality of CAMs further. Here, we refrain from using external datasets and additional supervision other than images from the target dataset when training DDPM.

As shown in Fig. 1, we utilize ViT for the classification backbone and a U-Net based network for DDPM while using only image-level supervision. Though DDPM captures semantic information with high-level locality, as shown in Fig. 1-(a), it needs to be semantically aligned with the semantics of the ViT structure. As such, we propose the Locality Fusion Cross Attention (LFCA) module to enable this alignment and use it to improve the locality of ViT-CAMs. Specifically, the LFCA module brings query tokens from the class-aligned patches of ViT and key/value tokens from the diffusion feature to fuse the different feature representations using Cross Attention. To guide the output tokens from LFCA, we apply a classification loss and further distill the knowledge from the ViT model using Kullback-Leibler Divergence (KLD) to successively align the semantics of LFCA with that of ViT. After sufficient alignment, Diffusion-CAMs generated from the output token of the LFCA is used to train the ViT-CAMs. Additionally, we introduce the Patch Affinity Consistency (PAC) that aims to train the ViT to robustly learn patch-to-patch relations. When images are noised and denoised using DDPM, the images undergo a human-imperceivable yet significant amount of degradation all the while semantic features persist. The denoised image can then be used as an augmented data sample as DDPM retains spatial and semantic consistency. As local semantics are unchanged during image degradation, patch affinities of the denoised image and the original image should be similar. PAC imposes a similarity constraint between the patch affinities of the original and degraded image to effectively guide the patch affinity, a parameter difficult to regularize using mere classification loss.

The main contributions of this paper can be summarized as follows:

- We propose the first-ever framework that successfully fuses a diffusion model to a ViT-based model for WSSS. Given the nature of the task, our method also only utilizes image-level supervision.
- We propose the Locality Fusion Cross Attention (LFCA) module to excavate the potential of the pretrained diffusion feature while aligning it to semantic classes from ViT.
- We design a simple yet effective Patch Affinity Consistency (PAC) that utilizes the denoised image as a spatially and semantically consistent augmentation sample to better guide the patch affinity used to refine CAMs.

Experimental results on the PASCAL VOC 2012 [15] and MS COCO [34] datasets validate the superiority of the proposed method by establishing a new state-of-the-art performance.

2 Related Works

2.1 Diffusion Models

The diffusion model [21, 43] is a probabilistic generative model that aims to approximate the distribution of real images. The diffusion model is learned to recover the image when the given image is corrupted with successive Gaussian noise. DDPM [21] modeled the forward diffusion process and proposed a training objective that reversed the diffusion process in a simplified manner. For faster inference in image generation, DDIM [44] proposed an implicit probabilistic model with a non-markovian chain. Dhariwal *et al.* [13] proposed a new architecture tailored for diffusion models, inspired by the observation that while GAN model architectures have seen significant progress, diffusion models have not received a commensurate level of attention. With the powerful model and modified objective, the diffusion model first surpasses the GAN model in terms of sample quality and diversity. In our work, we follow the process of the DDPM, and we adopted the U-Net based diffusion model proposed in [13].

2.2 Generative Model-based Image Segmentation

Similar to the purpose of WSSS to alleviate the annotation burden of semantic segmentation, there has been active research to utilize generative models for semantic segmentation. Before the emergence of DDPMs, most studies focused on leveraging the representation of GAN models to perform semantic segmentation. Tritrong *et al.* [49] proposed a one-shot semantic part segmentation framework that leverages GAN representation. Xu *et al.* [58] and Galeev *et al.* [17] show that GAN encodes the semantics of images in a simple manner and applying a linear transformation or constructing a lightweight decoder is sufficient to project the GAN feature into semantic segmentation map. With the emergence of diffusion models that surpass GANs [13] in image synthesis, much research [3, 4, 47, 53] has been aimed at utilizing Diffusion models for Semantic Segmentation. Baranchuk *et al.* [3] shows that diffusion models can be used for semantic segmentation while capturing high-level semantics valuable for other tasks. Rahman *et al.* [38, 53] proposed a diffusion model-based framework for medical image segmentation.

2.3 Weakly Supervised Semantic Segmentation

CAMs improvement. Most weakly-supervised semantic segmentation approaches using image-level labels utilize Class Activation Maps [67] to localize the object in an image. However, unrefined CAMs fail to discover the less-discriminative region while exhibiting imprecise object boundaries. To improve CAMs, various methods are proposed: leveraging cross-image semantic relations [16, 30, 45], attention mechanisms [37, 52, 55], splitting images into complementary patches [65], consistency between local-global view image [22], introducing extra-category [5, 60] and adversarial learning between the reconstructor and classifier [26]. Other

than these methods, Adversarial Erasing (AE) methods [25, 29, 46, 62, 66] based on erase and find mechanisms are introduced. In addition to expanding CAMs, contrastive learning-based methods [8, 57, 68] guide CAMs to have more precise object boundaries. With the advent of ViT, which shows promising localization capability, many WSSS works [18, 35, 40–42, 59, 60] thus employed ViT as a backbone for localize the object instead of CNN. To extract the attention map, MCTformer [59] extended a single class token with multiple class tokens and proposed a framework to extract and refine CAMs in a class-specific manner within ViT. Xu *et al.* [60] and Lin *et al.* [35] bring the Vision-Language (VL) model to enrich the class representation power of ViT. However, the language models that these methods use are trained on large external datasets and introduce additional language supervision, making them unfit for comparison with only image-level-based supervision methods. Though many ViT-based WSSS are proposed, to the best of our knowledge, there has yet to research that handles the lack of locality in ViT.

Refining CAMs. To further refine the quality of CAMs to serve as labels for semantic segmentation, [1, 2] leverage semantic affinities between nearby pixels and [7] uses boundary information. MARS [23] utilizes the feature from the unsupervised semantic segmentation model to generate de-biased pseudo labels. Mat-Label [51] brings the image-matting to the pseudo labels generation process in WSSS. BECO [39] proposes a co-training framework by mixing labels to improve pseudo labels in semantic boundaries. As these post-processing techniques rely on the quality of CAMs and can be integrated with **CAMs improvement** methodologies, our emphasis in this study is placed on the **CAMs improvement** phase rather than refining CAMs.

3 Method

3.1 Preliminaries

Denoising Diffusion Probabilistic Models. In our work, we employ diffusion model [21, 43] to provide well-structured high-level semantic information to the classifier that extracts the CAMs. DDPMs consist of a forward process $q(x_t|x_{t-1})$ that gradually corrupts the input data x_0 to Gaussian noise $x_T \sim \mathcal{N}(0, 1)$ in T steps, and a reverse (*i.e.* denoising) process $p_\theta(x_{t-1}|x_t)$ that predicts the noise in the forward diffusion process to get cleaner samples x_t .

As Ho *et al.* [21] empirically found that predicting the noise directly from the network $\epsilon_\theta(x_t, t)$ rather than to predict the mean $\mu_\theta(x_t, t)$ can generate more high-frequency details, we follow the prior work. The simplified training objective by Ho *et al.* [21] is,

$$\mathcal{L}_{simple} = E_{t, x_0, \epsilon} [||\epsilon - \epsilon_\theta(x_t, t)||^2]. \quad (1)$$

Therefore, during the training process of the diffusion model, no labels other than the images themselves are utilized. Additional mathematical details regarding DDPMs are in the *Supp. Materials*.

Vision Transformer for WSSS To feed the input image x_0 into ViT, the image is divided into $N \times N$ patches and subsequently embedded into patch tokens $\mathbf{T}_p \in \mathbb{R}^{N^2 \times D}$, where D represents the embedding dimension. We then concatenate C class tokens $\mathbf{T}_c \in \mathbb{R}^{C \times D}$ with the patch tokens $\mathbf{T}_p$ and add positional embeddings. The concatenated token input $\mathbf{T}^{in} \in \mathbb{R}^{(C+N^2) \times D}$ is forwarded to ViT, as in [59], and the output token $\mathbf{T}^{out} \in \mathbb{R}^{(C+N^2) \times D}$ is split into class token $\mathbf{T}_c^{out} \in \mathbb{R}^{C \times D}$ and patch token $\mathbf{T}_p^{out} \in \mathbb{R}^{N^2 \times D}$. By pooling $\mathbf{T}_c^{out}$, class prediction $\hat{y}_c$ is produced. After reshaping the patch token $\mathbf{T}_p^{out}$ and applying the convolutional layer with C channels, we can obtain CAMs $\mathbf{A} \in \mathbb{R}^{N \times N \times C}$ and class prediction $\hat{y}_p$ from ViT. Here, class prediction $\hat{y}_p$ is calculated as:

$$\hat{y}_p = \mathbf{TopK}(\mathbf{ReLU}(\dot{\mathbf{A}}), K) - \mathbf{TopK}(\mathbf{ReLU}(-\dot{\mathbf{A}}), K) \quad (2)$$

where $\dot{\mathbf{A}}$ and K are patch-level feature before $\mathbf{ReLU}$ applied for producing $\mathbf{A}$ and number of selection, respectively. $\mathbf{TopK}(\cdot)$ is the Top-K pooling operation that selects and averages K highest values along the spatial dimensions for each channel. Though the baseline [59] uses global average pooling to calculate class prediction, we modified the pooling process with Eq. 2. The classification loss $\mathcal{L}_{cls-vit} = \mathcal{L}_{cls}(\hat{y}_c, \mathbf{y}) + \mathcal{L}_{cls}(\hat{y}_p, \mathbf{y})$ is applied for supervising class predictions of ViT, $\hat{y}_c$ and $\hat{y}_p$, where $\mathcal{L}_{cls}(\hat{y}, \mathbf{y})$ denotes multi-label soft margin loss between the prediction $\hat{y}$ and the label $\mathbf{y}$. Furthermore, with ViT, we can extract the token-to-token attention map $\mathbf{A}_{t2t} \in \mathbb{R}^{(C+N^2) \times (C+N^2)}$. From this attention map, we can obtain the class-to-patch attention $\mathbf{A}_{c2p} \in \mathbb{R}^{C \times N^2}$ and the patch-to-patch attention $\mathbf{A}_{p2p} \in \mathbb{R}^{N^2 \times N^2}$. To aggregate global-local information, class-to-patch attentions from L layers of ViT are fused to generate class-specific attention map $\mathbf{A}_{att} = \sum_l^L \mathbf{A}_{c2p}^l$. Here, $\mathbf{A}_{att}$ is transposed and reshaped to form $\mathbb{R}^{N \times N \times C}$. Similarly, patch-to-patch attentions in L layers are fused and reshaped to produce $\mathbf{A}_{aff} \in \mathbb{R}^{N \times N \times N \times N}$, which represents affinity between patch tokens: $\mathbf{A}_{aff} = \sum_l^L \mathbf{A}_{p2p}^l$. Finally, to generate refined CAMs $\mathbf{A}_{ref} \in \mathbb{R}^{N \times N \times C}$, CAM $\mathbf{A}$ is refined by $\mathbf{A}_{att}$ and $\mathbf{A}_{aff}$:

$$\mathbf{A}_{ref}(i, j, c) = \sum_m^N \sum_n^N \mathbf{A}_{aff}(i, j, m, n) \cdot (\mathbf{A} \odot \mathbf{A}_{att})(m, n, c) \quad (3)$$

where $\odot$ represents element-wise multiplication and $\cdot$ represents multiplication.

3.2 Overview

As shown in Fig. 2, we propose two methods to propagate the capability of DDPMs to CAMs. Since our method utilizes pretrained DDPM without fine-tuning, a U-Net based model for DDPM is trained with the objective in Eq. 1. To align the semantically and locally well-clustered features of DDPM to semantic classes, we design the Locality Fusion Cross Attention (LFCA) method as shown in Fig. 3. Additionally, we leverage a denoised image $\tilde{x}_0$, made from iteratively adding noise to the original image x_0 and denoising it to the original

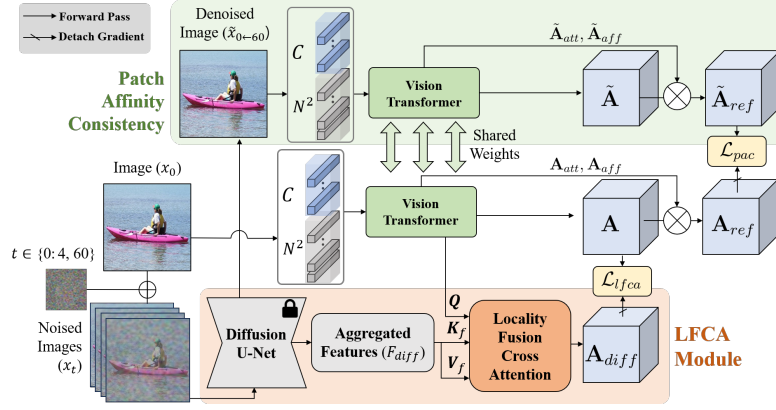


Fig. 2: Visualization of the proposed framework. The image x_0 is input into ViT with C class tokens and frozen DDPM with different time steps t . The $Key(\mathbf{K}_f)$ and $Value(\mathbf{V}_f)$ are made from the aggregated diffusion features. Meanwhile, $Query(\mathbf{Q})$ is extracted from patch tokens in later ViT layers. $\mathbf{Q}, \mathbf{K}_f, \mathbf{V}_f$ are utilized in Locality Fusion Cross Attention (LFCA) to generate semantically aligned feature that can be used to produce Diffusion-CAM $\mathbf{A}_f$. Concurrently, CAM $\mathbf{A}$ is created from output patch tokens of ViT. After semantically aligning the LFCA Module for a few training epochs, we apply an $L1$ loss between $\mathbf{A}$ and $\mathbf{A}_f$ to propagate diffusion-based semantic locality. Classification loss is omitted for simplicity.

step with DDPM, as an augmented sample. With this, we propose Patch Affinity Consistency(PAC) that checks for the consistent affinity between the patch tokens from clean image x_0 and denoised image $\tilde{x}_0$.

3.3 Locality Fusion Cross Attention

In LFCA, we aggregate diffusion features $F_t \in \mathbb{R}^{H_f \times W_f \times D_f}$ obtained by propagating noisy images x_t of various time steps t to the diffusion encoder (bottleneck features). Here, H_f and W_f are the height and width of the diffusion feature, and D_f denotes the feature dimension. The diffusion features from the different noisy images are concatenated and reduced, using a series of convolution and layer-norm operations, to obtain an aggregated diffusion feature $F_{diff} \in \mathbb{R}^{H_f \times W_f \times D_f}$. Though aggregated diffusion feature F_{diff} can provide meaningful locality information to the ViT, the diffusion feature is not yet semantically aligned with the ViT. To effectively align the semantics of the diffusion feature with that of ViT features, we fuse the query tokens from the ViT layers with the aggregated diffusion feature using the Cross-Attention mechanism. To compute Cross-Attention, we first extract the Query $\mathbf{Q} \in \mathbb{R}^{(C+N^2) \times D}$ from the ViT layers to provide class-aligned semantic information. To extract locally aligned information from the diffusion features, we first reshape the diffusion feature F_{diff} into $\mathbf{T}_f \in \mathbb{R}^{N_f^2 \times D_f}$ where $N_f^2 = H_f \times W_f$. And for the Key $\mathbf{K}_f \in \mathbb{R}^{N_f^2 \times D}$ and Value $\mathbf{V}_f \in \mathbb{R}^{N_f^2 \times D_f}$, they are extracted from aggregated diffusion feature F_{diff} . Then, we fuse the

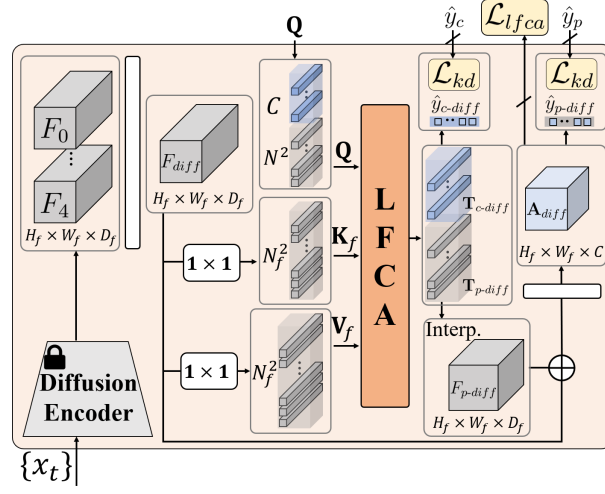


Fig. 3: Visualization of Locality Fusion Cross Attention module. The module uses Query Q tokens from the ViT layers to semantically align the diffusion features through cross-attention. KD Losses $\mathcal{L}_{kd}$ are computed for class predictions from both class tokens and patch tokens. The Diffusion-CAMs A_{diff} are used to guide ViT-CAMs to provide locality information.

feature according to the Cross-Attention mechanism:

$$\text{LFCA}(Q, K_f, V_f) = \text{softmax} \left(\frac{QK_f^\top}{\sqrt{D}} \right) V_f \quad (4)$$

We split the fused token $T_{lfca} \in \mathbb{R}^{(C+N^2) \times D_f}$ into class tokens $T_{c-diff} \in \mathbb{R}^{C \times D_f}$ and patch tokens $T_{p-diff} \in \mathbb{R}^{N^2 \times D_f}$ to apply of classification loss and to generate CAMs. T_{c-diff} is pooled in the D_f dimension to derive the class prediction $\hat{y}_{c-diff}$. The $\hat{y}_{p-diff}$ can be obtained as in Eq. 2. Simultaneously, T_{p-diff} is reshaped into a feature map of size $N \times N \times D_f$ and interpolated into a feature map $F_{p-diff} \in \mathbb{R}^{H_f \times W_f \times D_f}$. The summation of the feature map F_{p-diff} and the residual from F_{diff} is passed through a convolution layer to obtain the Diffusion-CAMs $A_{diff} \in \mathbb{R}^{H_f \times W_f \times C}$ and class prediction $\hat{y}_{p-diff}$. To supervise class predictions from diffusion feature, $\hat{y}_{c-diff}$ and $\hat{y}_{p-diff}$, we employed two different losses. Initially, multi-label soft margin loss $\mathcal{L}_{cls}(\cdot)$ is applied with image-level labels y . Furthermore, Kullback-Leibler (KL) Divergence is applied between class predictions to enable the diffusion features to comprehend and align with the probability distribution of ViT. Initially, $\hat{y}_c$ and $\hat{y}_{c-diff}$ are temperature-scaled with $\mathcal{T}$, and softmax function σ is applied to calculate probability distributions to be used in KL Divergence. We then define the KL Divergence loss as :

$$\mathcal{L}_{kl}(\hat{y}, \hat{y}_{diff}) = \mathcal{D}(\sigma(\hat{y}/\mathcal{T}) || \sigma(\hat{y}_{diff}/\mathcal{T})) \quad (5)$$

where $\mathcal{D}(P\|Q) = \sum_x P(x) \log(\frac{P(x)}{Q(x)})$ for probability distributions P and Q . Ultimately, to balance between the multi-label soft margin loss and KL Divergence loss for supervising $\hat{y}_{diff}$, the following is computed with balancing parameter α :

$$\mathcal{L}_{kd}(\hat{y}, \hat{y}_{diff}, \mathbf{y}) = \alpha \mathcal{L}_{cls}(\hat{y}_{diff}, \mathbf{y}) + \frac{(1-\alpha)}{\mathcal{T}^2} \mathcal{L}_{kl}(\hat{y}, \hat{y}_{diff}) \quad (6)$$

Here, $\mathcal{T}^2$ is introduced to correct for temperature scaling. The above equations can be applied to class predictions from both the class tokens and the patch tokens. The overall classification loss for training the LFCA module is as follows:

$$\mathcal{L}_{cls-diff} = \mathcal{L}_{kd}(\hat{y}_c, \hat{y}_{c-diff}, \mathbf{y}) + \mathcal{L}_{kd}(\hat{y}_p, \hat{y}_{p-diff}, \mathbf{y}) \quad (7)$$

Ultimately, the purpose of the LFCA module is to provide locality information from DDPM to ViT. However, as the LFCA module is not yet semantically aligned in the early stages of training, we supervise the ViT-CAMs $\mathbf{A}$ with Diffusion-CAMs $\mathbf{A}_{diff}$ only after a set amount of epochs has passed, using the following loss:

$$\mathcal{L}_{lfca} = |\mathbf{A} - \mathbf{A}_{diff}|_1, \quad (8)$$

where $|\cdot|$ denotes L1 loss. In Eq. 8, $\mathcal{L}_{lfca}$ does not backpropagate $\mathbf{A}_{diff}$, as Diffusion-CAMs is only used as supervision. Here, we resize $\mathbf{A}_{diff}$ through interpolation to match the size of $\mathbf{A}$.

3.4 Patch Affinity Consistency

Though the patch affinity $\mathbf{A}_{aff}$ can significantly improve the quality of ViT-CAMs($\mathbf{A}$) by using $\mathbf{A}_{aff}$ as refinement, patch affinity itself tends to increase the false positive activation as it has no spatial constraints and is only supervised by the classification loss. To improve patch affinity, we utilize the diffusion model as a form of data augmentation that ensures the affinity between patches remains relatively similar while maintaining spatial consistency. As opposed to the conventional use of DDPM to generate new images from random noise x_T , we use DDPM to recover the original image $\tilde{x}_0$ given a minimal yet sufficiently noised version (x_t) of the original image (x_0) for t steps. Unlike other data augmentations (*e.g.* transformation) that alter spatial consistency and affect patch-wise affinity, diffusion model denoises in a spatially consistent manner all the while preserving semantic content: the output blurry image still has a patch affinity similar to that of the original image. Though there exist other spatially consistent data augmentation techniques (*e.g.* Gaussian Blurring), they do not necessarily retain semantic consistency. Diffusion differs from these techniques in that it selectively denoises only within regions that share semantic alignment. This data augmentation approach ensures that information remains isolated between the regions with distinct semantics. Consequently, an image subjected to diffusion, denoted as $\tilde{x}_0$, retains its semantic details while exhibiting little changes in affinity, making it a viable candidate for affinity learning. To regulate the patch affinity using diffusion as a data augmentator, we pass both the original image

x_0 and denoised image $\tilde{x}_0$ through ViT. After passing through ViT, x_0 generates a refined CAM A_{ref} by Eq. 3, while $\tilde{x}_0$ undergoes ViT to derive the refined CAM $\tilde{A}_{ref}$. As in Eq. 3, patch-wise affinity $\mathbf{A}_{aff}$ and $\tilde{\mathbf{A}}_{aff}$ are applied to $\mathbf{A}_{ref}$ and $\tilde{\mathbf{A}}_{ref}$, respectively. Consequently, Patch Affinity Consistency (PAC) aims to train these affinities to be robust to the spatially and semantically consistent data augmentations using the following loss function:

$$\mathcal{L}_{pac} = |\mathbf{A}_{ref} - \tilde{\mathbf{A}}_{ref}|_1, \quad (9)$$

where $|\cdot|$ is L1 loss. PAC is a simple yet effective method for enhancing the resilience of patch affinity, an otherwise difficult parameter to regulate.

The final loss of the proposed framework is as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{cls-vit} + \mathcal{L}_{cls-diff} + \mathcal{L}_{lfca} + \lambda \mathcal{L}_{pac} \quad (10)$$

where λ is the hyperparameter for balancing between the loss terms.

4 Experiments

4.1 Experimental Settings

Datasets We evaluate our method on PASCAL VOC 2012 [15] and MS-COCO 2014 [34] datasets, which are two widely used multi-label datasets in WSSS. PASCAL VOC 2012 dataset contains 20 foreground object classes and one background class. It is composed of three subsets (*train*, *val*, *test*), and each contains 1464, 1449, and 1456 images, respectively. The *train* set of VOC 2012 dataset is augmented with SBD dataset [20] to 10582 images as in prior works [26, 59, 60, 62]. The MS-COCO 2014 dataset consists of 80 foreground object classes and one background class with two subsets (82k *train* images and 40k *val* images).

Evaluation Metric To evaluate the quality of CAMs and performance of semantic segmentation, we use mean Intersection over Union (mIoU) as in the previous works [26, 59]. When evaluating the quality of CAMs, in WSSS, the mIoU with the optimal threshold is reported. The quality of CAMs is commonly evaluated on the *train* set and the performance of the semantic segmentation model is evaluated on the *val* set. Though the ground-truth pixel-level labels on *val* set are reachable for both datasets and computed locally, the evaluation of the PASCAL VOC 2012 *test* set is done through the official website.

Implementation Details For a fair comparison, we employ DeiT-S pretrained on ImageNet [12] as the classification backbone, following the approach of previous ViT-based WSSS methods [18, 26, 59, 60]. The classification network undergoes training with the Adam optimizer for 60 epochs, using an initial learning rate of 5e-4. The data augmentation strategies are consistent with that of MCT-former [59] except that we do not employ color jittering. The features F_t for K_f and V_f are derived from the middle layer of the Unet-based DDPM model with $t \in \{0, 1, 2, 3, 4\}$. Drawing inspiration from [61], both positive and negative predictions are jointly incorporated to guide $\mathcal{L}_{kl}$ effectively. The temperature

Table 1: Ablation study on the PASCAL VOC 2012 *train* set. $\mathcal{L}_{cls-vit}^\dagger$ refers to the multi-label soft margin loss computed between the class predictions obtained through global average pooling (GAP) and class labels. $\mathcal{L}_{cls-diff}^-$ means the loss without KL Divergence loss $\mathcal{L}_{kl}$.

	$\mathcal{L}_{cls-vit}^\dagger$	$\mathcal{L}_{cls-vit}$	$\mathcal{L}_{cls-diff}^- + \mathcal{L}_{lfca}$	$\mathcal{L}_{cls-diff} + \mathcal{L}_{lfca}$	$\mathcal{L}_{pac}$	mIoU
MCTformer [59]	✓					63.3
Baseline		✓				65.7
(a)		✓	✓			66.8
(b)		✓			✓	67.4
(c)		✓			✓	67.3
(d)		✓	✓		✓	68.2
(e)	✓			✓	✓	67.9
(f)		✓		✓	✓	69.3

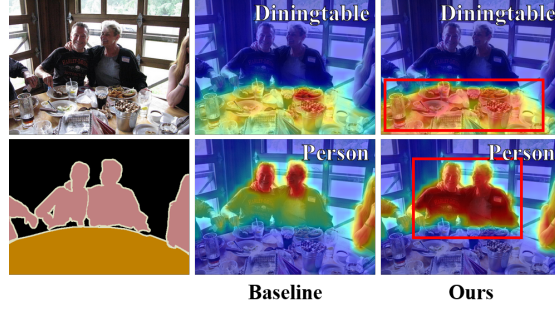


Fig. 4: Qualitative results of refined CAMs. The red box signifies uniformly activated areas corresponding to regions with the same semantic content.

parameter $\mathcal{T}$ for $\mathcal{L}_{kl}$ and balancing parameter α for $\mathcal{L}_{kd}$ are experimentally set to 5 and 0.2, respectively. $\mathcal{L}_{lfca}$ is applied from 10th epochs, which is selected based on classification accuracy. The multi-label soft margin loss is additionally applied to class predictions derived from the denoised image $\tilde{x}_0$. To balance the scale between loss terms, λ is set to 10. To ensure a fair comparison with previous CAMs improvement methods, we also applied the same post-processing model IRN [1] with prior works when generating pseudo labels. We also report the performance with different post-processing (PSA [2]). For the semantic segmentation model, we conduct experiments using two widely-used models: Deeplab V1 with a WideResNet38 backbone, and Deeplab V2 with a ResNet 101 backbone. Details regarding the architecture of the diffusion model are discussed in the *Supp. Materials*.

4.2 Ablation Studies

Component analysis To demonstrate the importance of each element in our proposed framework, we conducted ablation studies, as presented in Table 1. In our work, we adopt Min-Max K pooling as the baseline instead of the global

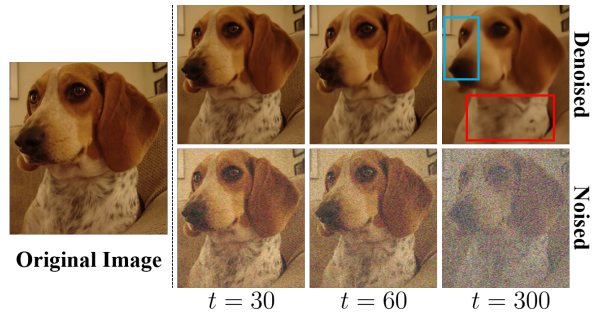


Fig. 5: Noised images (bottom) at different time step t and images denoised by Diffusion (top). The red boxes indicate significant deformations leading to changes in affinity. The blue boxes illustrate clear boundaries between regions with distinct semantic information.

average pooling (GAP) when computing the class prediction $\hat{y}_p$. Contrary to the experimental results of MCTformer [59] that show max pooling degrading CAMs **A** performance significantly, Min-Max K pooling showed a 2.4% gain in mIoU performance when compared with GAP given the same network structure. We can interpret this difference as coming from the use of negativity [62]. Additional discussions for Min-Max K pooling are in the *Supp. Material*. By training ViT-CAMs **A** using the Diffusion-CAMs $\mathbf{A}_{diff}$ obtained through the proposed LFCA, we achieved a performance improvement of 1.7% compared to the baseline (Table 1-(b)). We denote this as $\mathcal{L}_{cls-diff}$ in the table as the LFCA loss $\mathcal{L}_{lfca}$ is effective only after $\mathbf{A}_{diff}$ is sufficiently trained using $\mathcal{L}_{cls-diff}$. With the PAC that guides ViT to learn the relation between patches, we observed a performance improvement of 1.6% compared to the baseline as in result (c). Finally, when leveraging both $\mathcal{L}_{lfca}$ and $\mathcal{L}_{pac}$ for training, we achieved a remarkable 3.6% improvement over the baseline, demonstrating that the benefits brought by the proposed LFCA and PAC from diffusion model are independent and non-redundant. As evident from Fig. 4, it can be observed qualitatively that the locality has improved compared to the baseline, indicating enhanced performance. Though our method exhibits high performance without using the $\mathcal{L}_{kl}$, as shown in (d), distilling the knowledge of ViT helps align the semantic information of diffusion with ViT as shown by comparing (d) and (f). Lastly, our method also shows significant gains in mIoU performance (4.6%) when using GAP to compute class predictions, as shown in (e).

Effect of diffusion time steps In Fig. 5, we show the denoised image $\tilde{x}_0$ from various steps $t \in \{30, 60, 300\}$. In PAC, we use $t = 60$ as $\tilde{x}_{0 \leftarrow 60}$ is perceptually indistinguishable yet augmented enough for the network to discern. In the case of $t = 300$, the denoised image $\tilde{x}_{0 \leftarrow 300}$ is noticeably different from x_0 , to the point that the patch affinity may deviate too much from that of the original image. Experimentally, using denoised images with $t < 150$ showed similar mIoU performance ($67.3 \pm 0.3\%$), while higher steps showed degraded performance. This ablation is included in the *Supp. Material*.

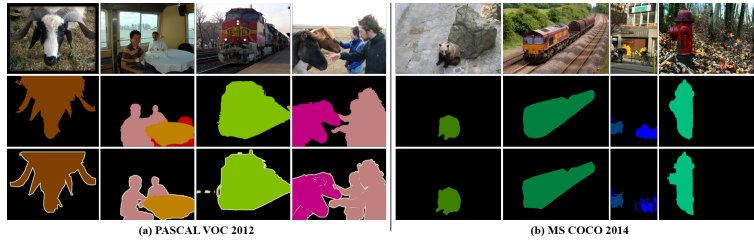


Fig. 6: Semantic segmentation visualization results on (a) VOC 2012 and (b) COCO 2014. From top to bottom: Image, Ours, GT.

Table 2: Comparison of our method with the state-of-the-art WSSS (multi-stage) methods. mIoU Performance is evaluated on PASCAL VOC 2012 *train* set in both CAMs and Mask-level. The backbone for each method is listed for an equal comparison. **bold** represents the best results. For a fair comparison, post-processing methods applied by each approach (PSA [2]/ IRN [1]) are also listed in the table. Here, (W)RN refers to (wide)ResNet.

Methods	Backbone	Seed	Post-processing	Mask
OC-CSE [25] <i>ICCV21</i>	WRN38	56.0	PSA	66.9
CPN [65] <i>ICCV21</i>	WRN38	57.4	PSA	67.8
PPC [14] <i>CVPR22</i>	WRN38	61.5	IRN	70.1
ReCAM [10] <i>CVPR22</i>	RN50	54.8	IRN	70.5
RIB [27] <i>NeurIPS21</i>	RN50	56.5	IRN	70.6
AEFT [62] <i>ECCV22</i>	WRN38	56.0	PSA	71.0
ACR [26] <i>CVPR23</i>	WRN38	60.3	IRN	72.3
Mat-Label [51] <i>ICCV23</i>	RN50	62.3	IRN	72.9
MCTformer [59] <i>CVPR22</i>	DeiT-S	61.7	PSA	69.1
LPCAM [9] <i>CVPR23</i>	DeiT-S	63.5	PSA	70.8
FPR [6] <i>ICCV23</i>	DeiT-S	63.8	-	-
USAGE [36] <i>ICCV23</i>	DeiT-S	67.7	PSA	72.8
Ours	DeiT-S	69.3	IRN	73.3
Ours	DeiT-S	69.3	PSA	74.3

4.3 Comparisons to State-of-The-Arts

Table 2 reports the performance in terms of CAMs (Seed) and pseudo ground-truth for semantic segmentation (Mask). The evaluation is conducted on the *train* set. Compared to the current state-of-the-art, the proposed method exhibits significantly superior performance in both seed and mask quality. As shown in Table 3, the semantic segmentation model trained with this high-quality pseudo ground truth achieves 73.9% and 73.7% in PASCAL VOC *val* and *test* set, respectively. Considering the quality of the pseudo labels is 73.3%, the semantic segmentation model shows much higher performance (0.6% $\uparrow$) than the labels. This result demonstrates that the labels generated through our method encode meaningful information for the segmentation model. BECO [39] interprets the low correlation between pseudo labels and the segmentation model often originating from false positive backgrounds or incomplete objects. The analysis concludes that noise in pseudo labels primarily occurs at semantic boundaries. Considering this aspect, we can infer that the pseudo labels obtained through our

Table 3: Comparison of mIoU (%) performance between our proposed method and existing WSSS works. Evaluation is conducted on both PASCAL VOC 2012 and MS-COCO 2014 datasets. We list the methods using only image-level classification labels in this table for a fair comparison. **Bold** represents the best results.

<i>Methods</i>	<i>Backbone</i>	<i>VOC val</i>	<i>VOC test</i>	<i>COCO val</i>
SIPE [8] _{CVPR22}	RN101	68.8	69.7	-
RIB [27] _{NeurIPS21}	RN101	68.3	68.6	43.8
FPR [6] _{ICCV23}	RN101	70.3	70.1	43.9
ReCAM [10] _{CVPR22}	RN101	68.5	68.4	42.9
USAGE [36] _{ICCV23}	RN101	-	-	44.3
Ours	RN101	71.8	72.4	46.6
OC-CSE [25] _{ICCV21}	WRN38	68.4	68.2	36.4
CPN [65] _{ICCV21}	WRN38	67.8	68.5	-
SIPE [8] _{CVPR22}	WRN38	-	-	43.6
AEFT [62] _{ECCV22}	WRN38	70.9	71.7	44.8
ACR [26] _{CVPR23}	WRN38	71.9	71.9	45.3
OCR [11] _{CVPR23}	WRN38	72.7	72.0	42.5
<i>ViT-based methods</i>				
MCT [59] _{CVPR22}	WRN38	71.9	71.6	42.0
LPCAM [9] _{CVPR23}	WRN38	72.6	72.4	42.8
USAGE [36] _{ICCV23}	WRN38	71.9	72.8	42.7
Ours	WRN38	73.9	73.7	45.5

method are more accurate in these semantic boundaries; accordingly, our model exhibits 2.3% higher precision than the baseline. Along with the experimental result on PASCAL VOC 2012, the semantic segmentation performance on the MS COCO 2014 dataset is also listed in Table 3. The CAMs from our model achieve 43.0% in 80K *train* set, and we obtained pseudo labels (46.1%) by applying IRN [1]. Here, the semantic segmentation model on MS COCO achieves 46.6%, outperforming the SoTA methods. Additional visualization results on CAMs and semantic segmentation prediction are in the *Supp. Materials*.

5 Conclusion

In this work, we take inspiration from the ability of diffusion models to capture high-level semantic information and devise a framework that uses a pretrained diffusion model to handle the lack of locality issue in ViT. By successfully projecting diffusion features to their corresponding semantic classes, our proposed Local Fusion Cross Attention (LFCA) module generates Diffusion-CAMs that provide locality guidance to ViT-CAMs. In addition, using diffusion-based data augmentation in Patch Affinity Consistency (PAC) improves CAM refinement by providing consistency for an otherwise difficult-to-regulate parameter. The proposed modules show their superiority and achieve state-of-the-art performance on both VOC and COCO datasets. Some limitations and potential future work of the proposed framework is that there exist some leeway in terms of design choice. For example, though we have found $t \in \{0, 1, 2, 3, 4\}$ to be apt, there still may be a better combination of time steps. Investigating these aspects of the framework opens up new possibilities for better diffusion-based WSSS performance.

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